Weakly-Supervised Chinese Review Sentiment Classification at Scale: Pseudo-Labeling, Specialist Decomposition, and Honest Out-of-Distribution Evaluation

Undergraduate Thesis — Natural Language Processing

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Code, out-of-fold predictions, and full experiment records: `https://github.com/shuakami/douban-sentiment`

Abstract

We study three-class sentiment classification (positive / neutral / negative) of Chinese Douban movie short reviews under a realistic low-resource setting: 600 human-verified semantic labels plus a large weakly-labelled corpus of 910,998 reviews with star ratings. Our study makes four contributions. First, we *measure* that star ratings are a poor proxy for semantic sentiment: on a 271,905-row confident pseudo-label pool, only ∼59% of 5-star reviews are predicted positive, and 3-star reviews skew negative rather than neutral. Second, we show that pseudo-labelling the corpus with a distilled ensemble and training a single compact student on 160K balanced rows reaches 87.5% accuracy with 84.8% neutral recall on the gold set — evidence that label quality at scale, not model capacity, is the binding constraint. Third, our deployed system couples a distilled MacBERT student, a frozen MegatronBERT-1.3B feature judge, and a corpus-trained neutral specialist, reaching **89.5% accuracy / 85.9 macro-F1** on 600-row out-of-fold evaluation — figures we show in Section 6.2 are inflated by second-order teacher leakage and duplicate-text fold leakage (leakage-corrected estimates: 83.3–84.0% three-class accuracy, 92.9% binary) — with neutral recall improved from 54.3% to 70.7%. Fourth, we construct **lab2000**, a 2,000-row cross-movie benchmark with two-pass LLM blind annotation (same annotator model, different few-shot views) and 479 human-adjudicated rows, and show that the gold-trained system generalises poorly to wild short reviews (65.0% three-class on the ood1000 *development* set) — but that adding only 1,000 actively-selected boundary examples lifts the SFT student by $+8.8$pp (62.6%→71.4%, McNemar $p = 4.0 \times 10^{-8}$). Scaling the same protocol with frontier-LLM pseudo-labels — up to ∼17K actively-selected examples — pushes the ensemble mix to **77.1% three-class** on the development set (a best-of-N estimate; 74.6–76.3% across five seeds, 77.1% seed-averaged by averaging the seeds' probability outputs before decoding; 5-fold cross-fit validated; significantly above the 1K-AL mix, McNemar $p = 3.9 \times 10^{-4}$), and lifts binary OOD to **92.3%**. We further report three empirical observations invisible at n=600: active-learning samples require a gold anchor to avoid prior collapse, hard labels may beat soft distillation under domain shift, and star ratings become the strongest signal precisely where text is most ambiguous. All negative results — more than twenty rejected architectures — are documented.

Keywords: *sentiment analysis; Chinese NLP; weak supervision; pseudo-labelling; knowledge distillation; pre-trained language models; calibration; out-of-distribution evaluation.*

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1 Introduction

Short reviews on Douban — China's most influential movie community — are terse, ironic, and frequently mix praise with criticism. A typical review reads *"特效无敌，但是剧情简直像一坨"* ("the special effects are unbeatable, but the plot is literally a pile of trash"). Such mixed-polarity text is precisely where three-class sentiment analysis (positive / neutral / negative) breaks down: the neutral class is both the most frequent source of ambiguity and the hardest to learn from few examples.

This project addresses a realistic low-resource scenario: a course assignment provides 600 reviews with human-verified semantic labels, while the platform additionally offers a corpus of nearly one million Douban comments carrying only star ratings (1–5). Two research questions organize the work:

1. **RQ1.** Can a small labelled set be amplified by a large weakly-labelled corpus?
2. **RQ2.** Can specialist decomposition — a dedicated neutral expert trained on corpus-scale pseudo-labels — repair the weakest class without degrading the rest?

Our answer is yes to both, but with an important honest caveat: the corpus amplifies *coverage*, not *knowledge*. Pseudo-labels are produced by our own models, so the flywheel saturates after one round; the true bottleneck is annotation noise on terse comments, which we quantify with a dedicated out-of-distribution study.

Contributions.

- A measured analysis showing star ratings are unreliable weak supervision for three-class sentiment (corpus-wide three-class agreement between pseudo-labels and ratings ≈48%).
- A pseudo-label flywheel producing a 271,905-row confident pool and a single compact student (87.5% acc / 84.8% neutral recall) trained with no gold labels in its training set.
- A deployed hybrid ensemble (89.5% acc / 85.9 macro-F1 on gold-OOF, inflated by teacher and duplicate-text leakage — corrected estimates 83.3–84.0% / 76.3–77.5; Section 6.2), neutral recall 70.7%, combining a distilled student, a frozen 1.3B-parameter judge, and a corpus-trained neutral specialist.
- A reproducible evaluation protocol (stratified 5-fold OOF, cross-fit fusion-weight selection, McNemar/bootstrap significance tests, temperature calibration) plus a leakage audit quantifying three inflation channels in the headline OOF numbers, and a 2,000-row cross-movie benchmark (lab2000) under which active-learning expansion adds +8.8pp over gold-only SFT ($p \approx 4 \times 10^{-8}$, development-set estimate) and scaling to 17K LLM labels reaches 77.1% three-class accuracy.
- A documented catalogue of more than twenty measured-but-rejected architectures, which we argue is itself a contribution: it maps the effective design space for this task.

2 Related Work

Sentiment analysis of Chinese text. Document-level sentiment classification has a long history, from feature-based methods (unigrams, character n-grams) to neural architectures (Kim, 2014; Socher et al., 2013; Joulin et al., 2017; Reimers and Gurevych, 2019); comprehensive surveys are given by Pang and Lee (2008) and, for Chinese specifically, by Tan and Zhang (2008). For Chinese, the absence of whitespace tokenization makes character-level features and pretrained language models especially effective. Douban reviews specifically have been used in earlier Chinese sentiment work — including weak-label pipelines combining dictionaries with rating-derived supervision (Fan et al., 2018) — but almost exclusively as a binary (positive/negative) task; the three-class setting with a genuine neutral/mixed class is less

studied and substantially harder. The difficulty of the neutral/mixed class is documented for document-level polarity classification more broadly (Martins et al., 2021).

Pre-trained language models. BERT (Devlin et al., 2019) and its robustly-optimised successor RoBERTa (Liu et al., 2019) established the pretrain-then-finetune paradigm. For Chinese, MacBERT (Cui et al., 2020) replaces the MLM objective with a corrected variant better suited to character-level inputs, ERNIE (Sun et al., 2019) integrates entity and phrase knowledge, StructBERT (Wang et al., 2020) injects phrase- and word-level structural information into pre-training, and SKEP (Tian et al., 2020) targets our task directly by pre-training with sentiment knowledge. Erlangshen-MegatronBERT (IDEA-CCNL, 2023) scales this family to 1.3B parameters with additional sentiment fine-tuning. We use all three families and compare them directly.

Weak supervision and self-training. When labelled data is scarce, distant supervision from noisy signals (star ratings here) and self-training on model predictions are standard remedies (Yarowsky, 1995; Lee, 2013). The Snorkel framework (Ratner et al., 2017) and the WRENCH benchmark (Zhang et al., 2021) formalise the combination of multiple weak label sources. The gap between star ratings and textual sentiment is itself long documented — ratings were already treated as imperfect proxies in early work (Pang and Lee, 2005) and rating–sentiment incongruence has been measured directly on review corpora (Abaiyan et al., 2026). Active learning (Settles, 2009) instead spends a small annotation budget on the most informative examples; our Stage-2 study uses margin/entropy sampling for exactly this purpose. Our work differs in that we *measure* the weakness of the dominant weak source (ratings) before relying on it, and we quantify where pseudo-labelling helps (neutral coverage) versus where it cannot (helping beyond the teacher's knowledge).

Knowledge distillation. Hinton et al.'s dark-knowledge distillation (Hinton et al., 2015) — surveyed comprehensively by Gou et al. (2021) — lets a small student mimic a large teacher's soft predictions. We distil a 7-channel teacher ensemble into a single MacBERT-base student and validate every ensemble decision with out-of-fold predictions. Knowledge distillation under distribution shift is itself an active research area (Zhang et al., 2025); our hard-vs-soft comparison in Section 8.2 is a small-scale instance of the same question.

Calibration. Neural classifiers are systematically over- or under-confident (Guo et al., 2017). We apply temperature scaling — a single-parameter post-hoc fix — and report expected calibration error before and after, because honest confidence display matters for a demo system.

LLMs as annotators and baselines. General-purpose LLMs are increasingly used for zero-/few-shot classification (Gilardi et al., 2023; Törnberg, 2023), for generating synthetic training data, and as annotators whose agreement with human labels is known to overstate quality on subjective tasks (Møller et al., 2024; Schroeder et al., 2025). We evaluate Qwen3 models (Qwen Team, 2025) at 0.6B/1.7B/4B scales both as zero-shot classifiers and as independent annotators in our OOD study, and we report a negative result on LLM-generated synthetic augmentation.

3 Task and Datasets

3.1 Task definition

Given a Douban movie short review x, predict $y \in \{\text{positive}, \text{neutral}, \text{negative}\}$. These reviews tend to be brief — the corpus median is only 24 characters — and a single line often mixes sarcasm, allusions to a

specific film, and conflicting evaluations. We count as neutral those reviews that carry no clear polarity: bare plot summaries, evenly balanced praise and criticism, and meta-commentary such as "seen it, no comment". Table 1 shows representative cases, including the mixed-polarity pattern that dominates the error analysis.

Table 1: Representative gold-set reviews (Chinese original with English gloss). "Mixed→Neg" denotes a praise-then-criticism review whose final polarity is negative.

Review	Gloss	Label	Len
"特效无敌，但是剧情简直像一坨。"	Effects unbeatable, plot a pile of trash	Neg	15
"演技、镜头、配乐都是顶级，现代经典。"	Acting, camera, score all top-tier; modern classic	Pos	18
"看在演员的份上值回票价，剧情很套路。"	Worth it for the cast; routine plot	Neu	17
"两个小时的生命浪费，想喜欢它都难。"	Two hours wasted; hard to like	Neg	17
"又是续集，还行吧。"	Another sequel; it's okay	Neu	9

3.2 Gold set (600 reviews)

The labelled evaluation set contains 600 reviews sampled from a single film page, each annotated with `label_semantic` (human-verified semantic sentiment) and `label_rule` (star-derived weak label). Class distribution: 316 positive / 92 neutral / 192 negative. Reviews are comparatively long (mean 110.8, median 80 characters). The neutral class is supported by only 92 examples — the smallest and hardest class. A text-level audit finds the 600 rows contain only 592 unique review texts (eight exact-duplicate pairs; the leakage audit in Section 6.2 discusses the consequence).

3.3 Weakly-labelled corpus (910,998 reviews)

The Moviedata-10M corpus (csuldw, 2019) — 4,428,475 raw Douban comments — yields 910,998 cleaned, deduplicated comments after our own cleaning, across 20,966 films and 249,105 users, with star rating (1–5), timestamp, and vote counts but *no* semantic labels. Review length: mean 40.6 / median 24 / p90 = 102 characters. Rating distribution: 1★ 7.2%, 2★ 12.6%, 3★ 34.0%, 4★ 31.0%, 5★ 15.2%.

3.4 Star ratings are not sentiment labels

Before using ratings as weak supervision, we tested them. Our deployed student labels all 910,998 corpus reviews; on the 271,905-row confident ($p \geq 0.7$) pool, the row-normalised rating-to-prediction crosstab is:

Table 2: Star rating vs. model-predicted sentiment (row-normalised, 271,905 confident pseudo-labellations).

Rating	Predicted positive	Predicted neutral	Predicted negative
1★	1.7%	7.3%	**91.0%**
2★	1.6%	13.7%	**84.7%**
3★	9.9%	33.3%	**56.8%**
4★	38.3%	32.7%	29.0%
5★	**58.8%**	23.1%	18.2%

Three patterns stand out. Only ∼59% of 5-star reviews are predicted positive by the model — a model prediction, not verified semantic positivity (a later 400-row human audit finds only 62.7% of 5-star reviews human-positive; Section 8.5); 3-star reviews skew *negative* (56.8%) rather than neutral; and even at 1–2⋆, ∼8–14% of reviews are predicted neutral. Rating-vs-model binary agreement on the confident pool is 81.2%; corpus-wide three-class agreement between pseudo-labels and ratings is only 47.8% (71.2% on the gold set). These patterns are consistent with the long-documented rating–sentiment gap (Pang and Lee, 2005; Abaiyan et al., 2026); our contribution is the quantification at corpus scale, not the phenomenon itself. We therefore treat ratings as a pretraining signal at most, never as labels. Figure 1 visualises the same table.

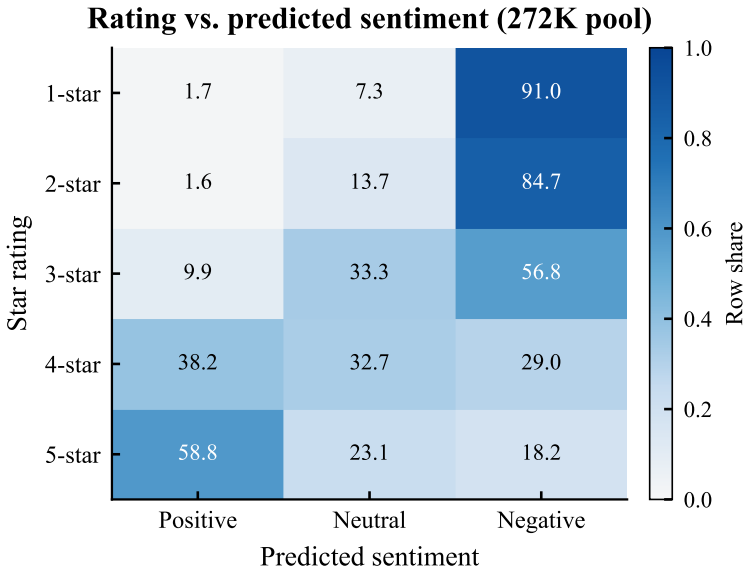


Figure 1: Star rating versus predicted sentiment on the 272K confident pseudo-label pool. The diagonal dominance is weak: ratings and model-predicted sentiment substantially disagree.

3.5 Stage-2 benchmark: lab2000

To measure — and then narrow — the gold-to-corpus generalisation gap documented in the lab500 pilot (Appendix B), we constructed **lab2000**: 2,000 corpus reviews in two disjoint, complementary subsets.

- **ood1000** (development set): 1,000 reviews sampled uniformly at random from the 911K corpus (gold-deduplicated, cross-movie). This set served as the *development* / model-selection set for the entire Stage-2 study: the AL scale (1K→17K), teacher choice (DeepSeek / local Qwen / hybrid), epoch count, seed subsets, and the 5-fold cross-fit fusion weights were all selected on it. It is therefore *not* an untouched test set; the only identifier-disjoint test is the 400-item blind human-annotation set human400 (Section 8.5; identifier-level disjointness is verified, and a text-level overlap audit is reported there).
- **al1000** (active-learning expansion): 1,000 reviews selected by margin/entropy sampling from the deployed student's predictions — deliberately the hardest, most ambiguous rows (neutral-heavy, low-confidence). This is a *training* set, not a test set.

Annotation protocol. Each row was labelled independently twice by two LLM annotation passes — the *same* annotator model with different few-shot example sets, blind to star ratings and to each other. This is single-model self-consistency, not two independent annotators; Section 8.3 discusses the implications for the agreement numbers. Inter-pass agreement was 86.7% (Cohen's $\kappa = 0.80$) on ood1000 and

79.8% ($\kappa = 0.67$) on al1000 — the lower agreement on al1000 is itself evidence that margin sampling selects intrinsically ambiguous rows. All 335 inter-pass disagreements plus 144 spot-checked agreements were adjudicated by the author (479 human-adjudicated rows total: 200 in ood1000, 279 in al1000); the remaining rows take the two-pass consensus label, so the released labels are mixed-provenance (ood1000: 200 human + 800 consensus; al1000: 279 human + 721 consensus). Final distributions: ood1000 = 398 positive / 327 neutral / 275 negative; al1000 = 473 / 369 / 158. A separate 200-row random subset of ood1000 was blind-labelled by a second human annotator to measure agreement with the released reference labels (Section 8.3).

One anchoring caveat applies to all 479 human-adjudicated rows: the adjudication interface displayed the deployed student model's prediction and confidence next to each item, so the adjudicator was not blind to model outputs — a documented source of anchoring bias in human-in-the-loop annotation (Schroeder et al., 2025). Adjudicated labels should be read as "human-verified with model-visible assistance", not as blind ground truth.

4 Methodology

4.1 Overview

The deployed system is a hybrid ensemble with three channels sharing a common backbone pass where possible:

1. **Distilled student**: MacBERT-base fine-tuned by distillation from a 7-channel teacher ensemble (Section 4.2).
2. **Frozen judge**: MegatronBERT-1.3B kept frozen; a small MLP head on its 4096-dimensional features (Section 4.3).
3. **Neutral specialist**: probability of neutrality computed from a corpus-trained detector blended with a judge-feature head (Section 4.5).

Figure 2 shows the data flow.

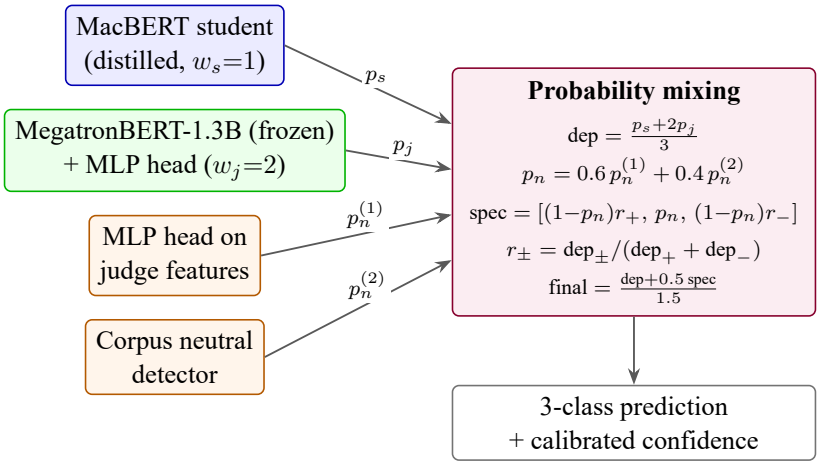


Figure 2: Deployed v3 architecture. $p_s, p_j, \text{dep}, \text{spec}, \text{final} \in \Delta^3$ are three-class probability vectors, whereas $p_n^{(1)}, p_n^{(2)}, p_n \in [0,1]$ are scalar neutral probabilities; w_s, w_j are fusion weights, not model parameters. $r_\pm$ denote the positive/negative shares of dep renormalised to sum to one. The two MegatronBERT-based heads share one frozen-backbone feature pass; the neutral specialist blends a judge-feature MLP with a corpus-trained MacBERT detector.

4.2 Teacher ensemble and distillation

A 7-channel teacher ensemble — TF-IDF logistic regression, frozen- backbone MLP heads on MacBERT, MegatronBERT-1.3B and Erlangshen variants, all trained on the 600 gold rows with honest out-of-fold predictions — reaches 86.8% three-class accuracy. We distil it into a single chinese-macbert-base student using hard-label cross-entropy plus soft-target KL divergence ($\alpha = 0.5$, temperature $\tau = 2$, 8 epochs, 5-fold OOF). The student alone reaches 80.2%/74.2%; blended 1:2 with the judge it reaches 87.8%/81.8%. Its role in the deployed system is to provide a cheap, well-calibrated channel that also serves as the pseudo-labeller for the corpus (Section 4.4).

4.3 Frozen judge

`IDEA-CCNL/Erlangshen-MegatronBert-1.3B-Sentiment` provides a 1.3B-parameter backbone pretrained with a sentiment-aware objective. We keep it frozen and train a small MLP head on its 4096-d features over the gold set. The deployed mixture weights the judge double the student:

$$\text{dep} = \tfrac{1}{3}\, p_{\text{student}} + \tfrac{2}{3}\, p_{\text{judge}}.$$

The two MLP heads (binary and three-class) share a single backbone feature pass via a cache, so the judge costs one forward pass per request regardless of task.

4.4 Pseudo-label flywheel

The deployed student labels all 910,998 corpus reviews. The confident subset ($\max p \geq 0.7$) contains 271,905 rows: 56.5K positive, 68.6K neutral, 146.8K negative. This pool is the raw material for every corpus-scale model in Section 4.6. A second flywheel round was tested and rejected: re-labelling with round-2 models produces degenerate probabilities (mean confidence 0.986, memorisation), so exactly one round is used. This saturation is consistent with the confirmation-bias dynamics documented for self-training, pseudo-labelling, and consistency-training methods (Arazo et al., 2020; Xie et al., 2020a,b; Sohn et al., 2020): once the pseudo-labels encode the teacher's own decision boundary, additional rounds amplify self-consistency rather than add information.

4.5 Neutral specialist

The neutral class is the weakest link (92 gold examples). We attack it on two fronts.

Corpus-trained detector. A MacBERT binary classifier is trained on a 115,000-row pseudo-labelled subset of the confident pool, sampled with a per-class cap of 60K rows per side, teaching the boundary between "genuinely neutral" and "polarised" reviews.

Specialist blend. The probability of neutrality combines a judge-feature MLP head and the corpus detector:

$$p_n = 0.6 \cdot P_{\text{neu}}(\text{MLP on judge feats}) + 0.4 \cdot P_{\text{neu}}(\text{corpus detector}).$$

From the dependency prediction $\text{dep} = (d_+, d_n, d_-)$ we take the positive/negative shares renormalised to sum to one, $r_+ = d_+/(d_+ + d_-)$ and $r_- = d_-/(d_+ + d_-)$, replace the neutral mass by the specialist estimate p_n, and form the specialist distribution

$$\text{spec} = \big[(1 - p_n)\, r_+,\ p_n,\ (1 - p_n)\, r_-\big],$$

then blend it into the final output:

$$\text{final} = \frac{\text{dep} + 0.5\,\text{spec}}{1.5}.$$

Every weight ($0.6/0.4$ detector mix, 0.5 blend) was selected by 20 random half-splits of the gold set and is stable across $w \in [0.3, 0.6]$.

4.6 Corpus-scale students

Using the 271,905-row pool, we trained corpus-only models (zero gold training) to measure how far pseudo-labels alone can go: a balanced 160K-row three-class student, a 195K-row binary student, a 115K-row neutral detector v1, and a hard-negative-mined v2 (rejected). These models double as baselines and as candidates for ensemble channels; Section 6 reports which (if any) survived.

4.7 Calibration

The raw ensemble is strongly under-confident (ECE = 0.251: predictions at 0.7–0.8 confidence are 98.9% accurate). We fit a single temperature $T = 0.37$ by NLL on the gold-600 OOF predictions, reducing ECE to ≈0.04 without changing any predicted label or accuracy. T is applied at inference so displayed confidence is honest. Because T is both fitted and evaluated on the same gold-600 OOF predictions, this is in-sample calibration evidence: it does not by itself establish calibration on the deployment distribution, and we do not claim deployment-grade calibration from it alone. Calibration also transfers better after AL training: on ood1000 the scaled AL mixes reach ECE 0.027–0.063 vs. v3's 0.159 at $T{=}1$ — pseudo-labelled boundary data sharpens confidence, not just accuracy.

4.8 Deployment

A FastAPI backend serves predictions; a React/Vite frontend provides the demo UI. An exact-match lexicon guard intercepts ultra-short slang before any GPU work. Warm latency is ≈0.1 s per review on an AMD RX 9070 GRE (ROCm); cold start ≈14 s.

5 Experimental Setup

Evaluation protocol. All headline numbers are **stratified 5-fold out-of-fold** predictions on the 600-row gold set: every model predicts only rows it never trained on. Corpus-scale models are additionally evaluated on a 4,000-row sample of fresh corpus reviews scored against rating-implied labels; we call this *rating-proxy agreement* rather than semantic out-of-distribution accuracy.

Metrics. Accuracy and macro-F1 (3-class), per-class precision/recall/F1, neutral recall specifically, and ECE for calibration. Significance is tested with McNemar's test on discordant pairs and percentile bootstrap on ΔF1; robustness is checked with 20 random half-splits.

Baselines.

- TF-IDF (char n-gram) + logistic regression, gold-trained.
- StructBERT sentiment classifier (Ali DAMO), zero-shot binary; `damo/nlp_structbert_sentiment-classification_chinese-base`
- Qwen3-0.6B / 1.7B / 4B chat models, zero-shot and 3-shot, scored by label log-probabilities.

6 Results

6.1 Main results

Table 3: Gold-600 results. "Deployed" rows are the production system (5-fold out-of-fold); StructBERT and Qwen rows are zero-shot external baselines — StructBERT is binary-only and scored on the 508 non-neutral base rows. The deployed-v3 OOF figures are inflated by second-order teacher leakage, duplicate-text fold leakage, and by tuning on the evaluation set; leakage-corrected estimates are reported in Section 6.2. "—" = metric not reported.

System	Task	Acc	Macro-F1
TF-IDF + LR	3-class	0.7233	0.6188
TF-IDF + LR	binary	0.8406	0.8232
StructBERT (DAMO)	binary	0.8661	0.8459
Qwen3-4B zero-shot	3-class	0.7833	0.7348
Qwen3-1.7B zero-shot	3-class	0.7067	0.5739
Qwen3-0.6B zero-shot	3-class	0.6817	0.4764
Teacher ensemble (7ch)	3-class	0.8680	—
Deployed v2	3-class	0.8900	0.8432
Deployed v3	**3-class**	**0.8950**	**0.8594**
Deployed v3	binary	0.9587	0.9558

The deployed system beats the TF-IDF baseline by +17.2pp accuracy and zero-shot Qwen3-4B by +11.2pp. (3-shot variants were also measured: few-shot helps slightly at 4B — 78.8% vs. 78.3% — but collapses at 1.7B, 51.0%, due to negative-class bias; we report zero-shot as the primary baseline.) Per-class metrics (3-class): positive P/R/F1 = 0.909/0.943/0.926; neutral = 0.765/0.707/0.735; negative = 0.931/0.906/0.918. Neutral recall improved 54.3% → 63.0% (v2) → 70.7% (v3) across expert iterations. Figure 3 shows the v3 confusion matrix: remaining errors concentrate in neutral↔polar confusions (47/63).

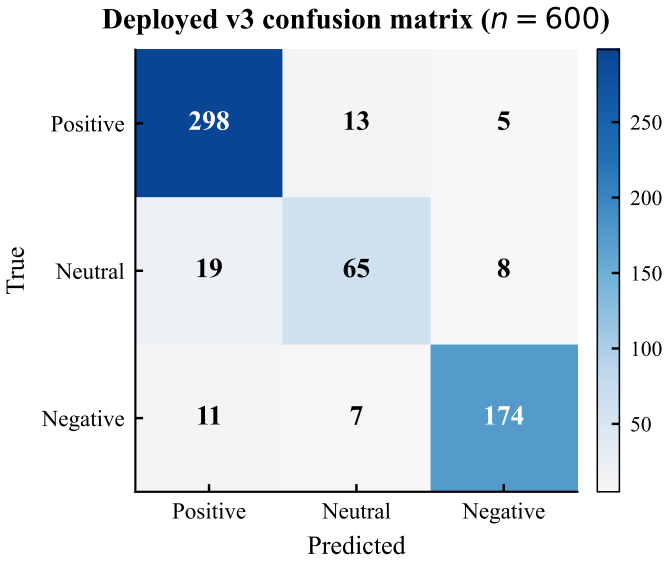


Figure 3: Deployed v3 confusion matrix on the 600-row gold set (out-of-fold). Row sums match the class distribution (316/92/192).

6.2 Leakage audit of the gold-OOF numbers

A post-hoc audit of the training and evaluation chain shows that the 89.5%/85.9% (three-class) and 95.87%/95.58% (binary) figures in Table 3 are *not* strict full-pipeline out-of-fold estimates. Three leakage channels inflate them, and every ensemble decision was tuned on the same 600 rows. We report the audit because the honest numbers are the ones a reader should quote.

Leak B: second-order teacher leakage. Every non-zero-weight teacher channel in the distillation ensemble was itself trained on gold rows overlapping the student's evaluation folds, and the $\alpha = 0.5$ soft targets transmit the teachers' fold-specific biases into the student — a second-order form of target leakage (Kapoor and Narayanan, 2023; Sainz et al., 2023). Re-running the identical pipeline with hard labels only ($\alpha = 1.0$, same folds and hyperparameters) gives **84.0% accuracy / 77.5 macro-F1** (three-class, corpus channel kept) and **83.3% / 76.3** (three-class, corpus channel removed), versus **92.9% / 92.4** (binary). Leak B alone therefore accounts for roughly $+5.5$pp three-class and $+3.0$pp binary — far above the $\sim$1pp a single student channel would contribute, because the effect compounds across the ensemble.

Leak A: transitive pseudo-label contamination. The deployed corpus neutral detector was trained on pseudo-labels that ultimately descend from a teacher fitted on all 600 gold rows, so its labels leak gold information into the "corpus" channel. Removing that channel under $\alpha = 1.0$ costs a further $\approx$0.7pp (84.0$\rightarrow$83.3), quantifying Leak A. The same channel was trained on 13.0% of the ood1000 rows and 13.2% of the fresh500 rows (Section 9), which is why those benchmark numbers carry a row-level overlap caveat as well.

Leak C: duplicate-text fold leakage. A text-level audit of the gold set finds 600 rows but only 592 unique review texts: eight exact-duplicate pairs (kept$\leftrightarrow$dropped review ids 63$\leftrightarrow$101, 64$\leftrightarrow$109, 68$\leftrightarrow$110, 71$\leftrightarrow$115, 75$\leftrightarrow$117, 82$\leftrightarrow$118, 99$\leftrightarrow$104, and 100$\leftrightarrow$108; seven of the eight pairs are neutral or negative). All reported gold-OOF numbers were produced on the un-deduplicated 600-row file with ungrouped stratified 5-fold splits, so a duplicate's twin can land in a training fold while its copy is scored — expected to affect $\sim$6–7 of the 8 pairs. This *duplicate-text fold leakage* inflates the gold-OOF figures by an estimated $\sim$1–2pp, concentrated in the neutral/negative classes where the duplicates sit. A deduplicated 592-row variant was constructed for verification (`data/reviews_600_dedup592.jsonl`) and the full pipeline retrained on it with dedup-aligned teacher targets (`scripts/dedup_retrain.sh`, `scripts/eval_dedup_pipeline.py`): the leakage-reduced α=1.0 estimates move to **82.6–83.1%** three-class accuracy / 74.8–75.9 macro-F1 (from 83.3–84.0 / 76.3–77.5 on the 600 rows) and **91.5%** / 90.8 binary (from 92.9 / 92.4), consistent with the $\sim$1pp inflation estimate; the deployed-weight (α=0.5) variant drops more (86.5% vs. 89.5%), as expected since teacher soft targets also encoded the duplicated rows.

Tuning on the evaluation set. Fusion weights, ensemble selection, and distillation hyperparameters were all chosen on the same 600 gold rows, so even the corrected numbers carry a best-of-N optimism we do not quantify. We therefore quote the deduplicated α=1.0 rerun as the honest headline: **three-class 82.6–83.1% accuracy / 74.8–75.9 macro-F1; binary 91.5% / 90.8** (83.3–84.0% / 76.3–77.5 and 92.9% / 92.4 on the 600 rows). These are leakage-reduced OOF ablation estimates: single-seed runs (α=1.0 was not re-tuned) with the deployed fusion weights and ensemble selection kept frozen rather than re-selected on the corrected channels, so they are not fully selection-independent. The deployed system itself is unchanged — this audit concerns the evaluation, not the artifact. Data-contamination and leakage taxonomies motivating this audit follow Sainz et al. (2023); Kapoor and Narayanan (2023);

Balloccu et al. (2024); the underlying mechanism is label noise in the pseudo-label pool (Northcutt et al., 2021).

6.3 Corpus-scale single models

Table 4: Corpus-trained models evaluated on gold-600 (zero gold training unless noted). "—" = metric not reported.

Model	Train data	Acc	Macro-F1	Neu recall
TF-IDF + star-rating labels	300K rating	0.6183	0.5369	—
TF-IDF + pseudo labels	200K pseudo	0.8117	0.7766	—
MacBERT binary (rating)	115K rating	0.9232	0.9184	—
MacBERT neutral v1	115K pseudo	0.9017	0.8328	—
MacBERT binary pseudo	195K pseudo	0.9488	0.9455	—
MacBERT 3-class pseudo	160K pseudo	0.8750	0.8463	**0.848**

The pseudo-label 3-class student alone reaches 87.5%/84.6 with 84.8% neutral recall — nearly matching the deployed ensemble. This is the central empirical finding: **label quality at scale dominates model capacity**. The same MacBERT architecture trained on star ratings instead of pseudo-labels collapses on the three-class task.

6.4 Pseudo-label scaling

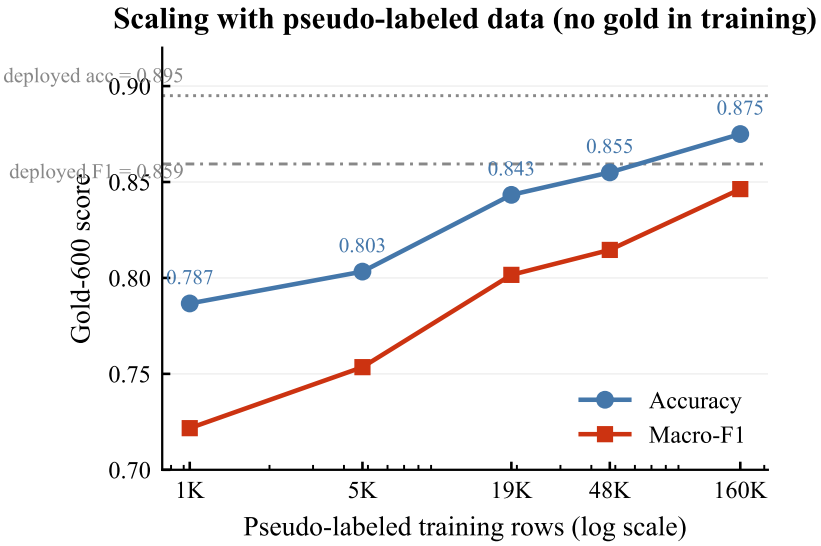


Figure 4: Accuracy and macro-F1 of corpus-trained pseudo-label students on gold-600 versus training-set size (log scale). *Zero gold training*: gold-600 is the held-out evaluation set only and is never used for training. The dashed reference is the deployed ensemble (accuracy 0.895, macro-F1 0.859). The y-axis starts at 0.70 to highlight the performance differences. Performance grows monotonically and is still rising at 160K rows.

Table 5 and Figure 4 show a monotone scaling curve: ∼1K rows → 78.8%, 5K → 80.3%, 19K → 84.3%, 48K → 85.5%, 160K → 87.5%. The curve has not saturated — corpus scale, not model size, is the active ingredient within the ranges tested.

Table 5: Pseudo-label scaling (gold-600 eval; no gold rows in training). "—" = metric not reported.

Pseudo rows	Acc	Macro-F1	Neutral recall
∼1K	0.7867	0.7217	—
∼5K	0.8033	0.7535	—
∼19K	0.8433	0.8016	—
∼48K	0.8550	0.8146	—
∼160K	0.8750	0.8463	0.848

6.5 Cross-domain rating-proxy evaluation

On 4,000 fresh corpus reviews scored against rating-implied labels, the round-3 student (trained on v3-teacher pseudo-labels) reaches 0.580 three-class / **0.837 binary** agreement versus rating, versus 0.552/0.803 for the round-1 student. These numbers measure *rating-proxy agreement* — agreement with star-rating-derived proxy labels — not semantic out-of-distribution accuracy, and should not be read as the latter. Ensemble-derived labels transfer slightly better out-of-domain even where gold scores are equal.

7 Analysis

7.1 Calibration

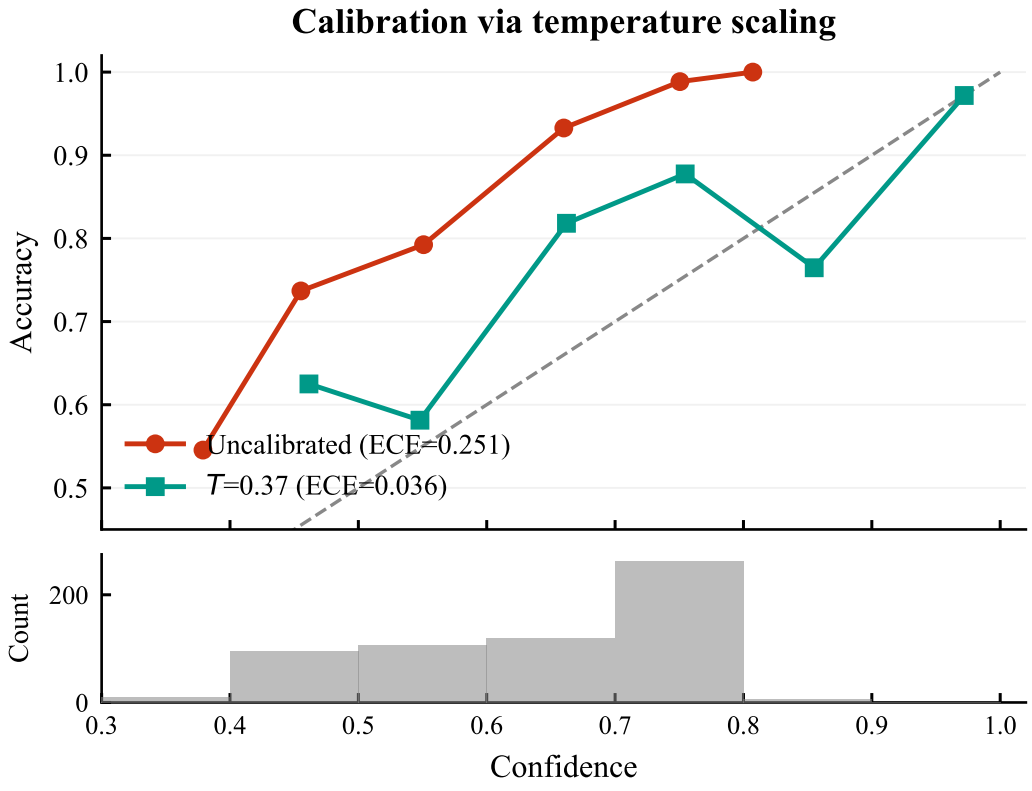


Figure 5: Reliability diagram before and after temperature scaling ($T = 0.37$, fitted by NLL on the gold-600 OOF predictions; all quantities are computed on the same 600 out-of-fold rows). The lower panel shows the sample count per equal-width confidence bin; the high-confidence region is sparse, so local fluctuations are expected even though the global ECE improves substantially. The accuracy axis is truncated at 0.45 to show the pre-scaling miscalibration.

The raw ensemble is under-confident (ECE 0.251); temperature scaling with $T = 0.37$ reduces ECE to ≈0.04 (Figure 5) without altering any predicted label or accuracy — it only reshapes the confidence distribution. One caveat: ECE averages over all bins, so a handful of sparse bins can still deviate noticeably even though the aggregate calibration improves substantially. We stress that T was both fitted and evaluated on the same 600 gold-OOF rows, so the ECE reduction is in-sample calibration evidence and does not by itself establish calibration on the deployment distribution; the OOD ECE figures reported for the AL mixes in Section 8 provide the out-of-domain check. This matters for the demo: displayed confidence now tracks empirical accuracy on the evaluation set.

7.2 Statistical significance

v3 vs. v2: McNemar's test (McNemar, 1947) on discordant pairs gives $p = 0.65$ (11 vs. 8); a paired bootstrap over resampled evaluation rows (Efron and Tibshirani, 1993) gives a 95% CI on ΔF1 of $[-0.5, +3.9]$pp. Following standard practice for comparing classifiers (Dietterich, 1998; Demšar, 2006; Dror et al., 2018), we treat differences inside the CI as unresolved. The point estimate (+1.6pp F1, +7.7pp neutral recall) is consistent across 20 random half-splits for every detector weight $w \in [0.3, 0.6]$. We report this honestly: specialist decomposition yielded a positive point estimate and improved neutral-class recall, although the overall Macro-F1 improvement was not statistically significant; at $n = 600$ the gold set cannot resolve differences below ∼1pp.

7.3 Error analysis

Of the 63 errors that remain, 47 involve neutral↔polar confusions, and most sit on reviews that even human annotators find ambiguous — mean confidence on errors is 0.51 versus 0.66 on correct predictions. Reviews that state an explicit score ("7.6/10", "三星半"; $n = 56$) behave quite differently: only 5.4% of them are misclassified, compared with 11.0% elsewhere. The model clearly exploits overt rating cues when they are present; just 6 of the 56 explicit scores even disagree with the semantic label.

7.4 Rejected architectures

More than twenty alternative designs were implemented, measured, and rejected; the deployed v3 is the measured optimum. Among the documented rejects: a neutral-gate cascade (its threshold overfit — split-half validation collapses from 87.6% to 86%), corpus-trained MLP heads as judge replacement, student-probability averaging, self-distillation round 2, stacked logistic meta-learner (0.837/0.794 — channels too correlated for 600 samples), class-weighted and max-len-128 students, Erlangshen 110M/330M students (pathologically slow on ROCm), neutral-detector v2 (worse in every ensemble combination), r3 self-training on v3-teacher labels (student 0.8767/0.8434, all mixes below v3), gold-mixed corpus training (circular gold evaluation — flagged, not cited as a result), Qwen3-4B synthetic augmentation (poisoned training: 79.2% standalone), and, in the Stage-2 OOD setting: AL-only training (56.0% — margin sampling without a gold anchor collapses the class prior) and gold+al soft-label distillation (71.0% vs. 71.4% for hard-label SFT — the narrow-domain teacher's OOD bias transfers through soft targets).

7.5 Comparison with a commercial training platform

For completeness we ran a preliminary comparison against a commercial NLP training platform (Alibaba Cloud PAI, StructBERT classifier, same 600-row task; our upload files are in `data/aliyun/`). Based only on the platform's published report page — we did not retain raw evaluation exports, so the denominators and split cannot be independently verified — the reported metrics indicate 89.83% accuracy, while the per-class metrics on the same page imply only 86.05% accuracy / 83.23% macro-F1 on

an effective validation set of **43 rows** (the three headline numbers correspond to denominators 59/60/119 and cannot be mutually reconstructed; the support-weighted recomputation also fails to close). A Wilson interval on 43 rows spans [72.7%, 93.4%] — the reported sample size cannot distinguish 86% from 90%. Reported training-set self-evaluation reached 99.02% ($n = 511$), a ∼13pp generalisation gap. These figures are indicative only and should not be treated as verified benchmark results. By contrast, our headline 89.5%/85.9 (pre-audit gold-OOF figures; Section 6.2) is computed on 600 out-of-fold rows with per-row predictions available for audit, and the two evaluations use different test sets — this is a protocol-quality observation, not a like-for-like accuracy claim.

8 Stage-2: The lab2000 OOD and Active-Learning Study

The lab500 pilot showed that gold-600 performance does not transfer to terse corpus reviews and that LLM reference labels are too noisy to serve as ground truth. We therefore built lab2000 (Section 3.5) and ran a full exposure-and-mitigation study. (The lab500 protocol and numbers are retained in Appendix B for completeness.)

8.1 Exposing the OOD gap

Table 6 reports every system on the ood1000 *development* set — the same set on which all Stage-2 selection decisions were made (Section 3.5), so these are development estimates, not untouched-test results. The deployed v3 — trained exclusively on gold-600 — drops from 89.5% in-domain to **65.0%** three-class accuracy (binary pos/neg: 85.0%). The gap is concentrated exactly where the lab500 pilot predicted: the neutral boundary.

Table 6: ood1000 (development set) results. "SFT" = MacBERT fine-tuned on the listed training data; "mix" = probability average of v3 and the SFT model, with the fusion weight selected by stratified 5-fold cross-validation on ood1000 — each prediction uses weights tuned on the other folds, so no reported row's label influences its own mix weight. Because the set itself drove all Stage-2 selection decisions, these are development estimates subject to best-of-N optimism (Section 9). Binary evaluation excludes neutral examples. "—" = metric not reported for this run. lab2000 labels are mixed-provenance: 479 human-adjudicated rows (200 ood + 279 al) plus two-pass LLM consensus labels for the rest (Section 3.5).

System	Training data	3-class acc / F1	Binary acc / F1
Deployed v3 (ensemble)	gold-600	0.650 / 0.644	0.850 / 0.849
SFT student	gold-600	0.626 / 0.611	0.832 / 0.831
SFT student	al1000 only	0.560 / 0.519	0.577 / 0.512
SFT student	gold-600+al1000	0.714 / 0.710	0.880 / 0.875
SFT student (distilled)	gold-600+al1000	0.710 / —	0.825 / —
mix (deployed cand.)	gold-600+al1000	**0.720 / 0.717**	**0.878 / 0.876**

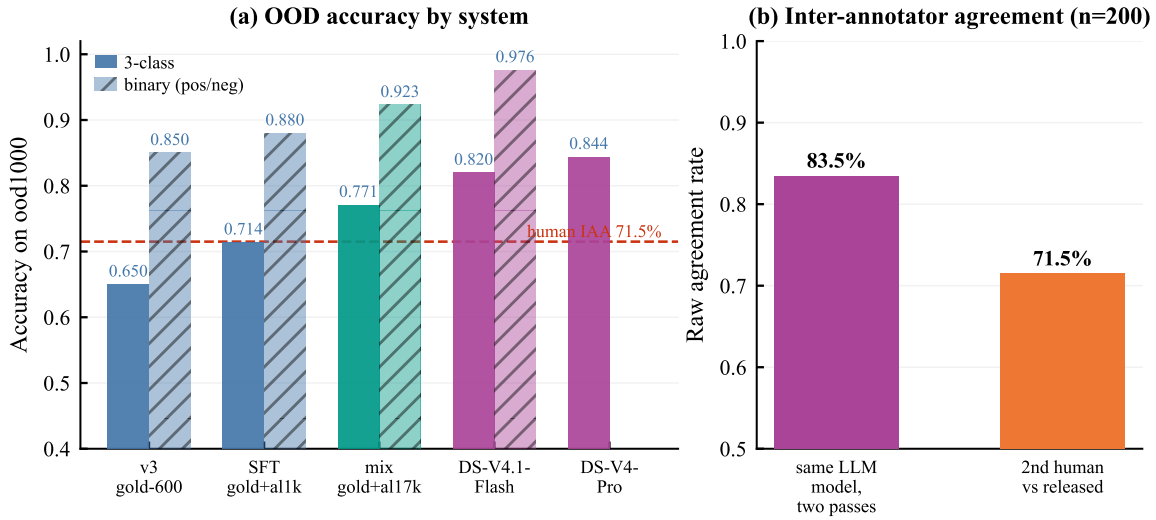


Figure 6: (a) ood1000 accuracy by system; the first three groups are our local models and the last two are off-the-shelf DeepSeek LLMs (the local Qwen3-4B baseline from Table 7 is omitted for space); the dashed line marks the measured 71.5% second-annotator-vs-mixed-reference agreement on the 200-item subset (Section 8.3). (b) Raw pairwise agreement on the same 200-row subset: two passes of the same LLM annotator model vs. a second human annotator scored against the released (mixed-provenance) labels. Raw agreement rate is reported here, not Cohen's κ; higher LLM self-agreement reflects shared inductive bias (same model) and literal rubric following rather than superior annotation quality.

Adding the 1,000 actively-selected AL rows (279 human-adjudicated + 721 LLM-consensus labels) to the training set lifts three-class OOD accuracy by **+8.8pp** (62.6%→71.4%, McNemar $p = 4.0 \times 10^{-8}$) — a gain that was statistically unresolvable at n=600 ($p = 0.65$ for v3-vs-v2). The ensemble mix reaches **72.0% / 0.717** three-class and **87.8% / 0.876** binary under cross-fit weight selection. Cross-fitting keeps each row's weight honest with respect to its own label, but the *set* itself drove every Stage-2 selection decision — AL scale, teacher, epochs, seed subsets, and these very weights — so we report all ood1000 numbers as development estimates carrying an estimated $+0.3$–1.0pp best-of-N optimism (Section 9).

8.2 Three empirical observations invisible at n=600

F1: Active learning needs an anchor. Training on al1000 *alone* collapses to 56.0% — worse than gold-600 alone (62.6%). Margin sampling selects boundary cases and systematically distorts the class prior; without the gold anchor the model loses its reference for ordinary high-confidence reviews. Gold and AL are complements, not substitutes.

F2: Under domain shift, hard labels may beat soft distillation. The gold+al teacher's soft targets transfer at 71.0% three-class vs. 71.4% for hard-label SFT — a 0.4pp gap that is within noise at n=600, so we report it as directional only. The direction is nonetheless consistent with the documented tendency of teachers to transmit their own biases through soft probabilities under distribution shift (Zhang et al., 2025).

F3: Ratings beat text exactly where text fails. On the al1000 ambiguity set, the star-rating rule reaches 64.3% agreement with the adjudicated labels while the text-only deployed model reaches ≈47%. When review text is maximally ambiguous — the regime margin sampling selects for — the author's star rating is the only remaining attitude signal. This closes the loop with Section 3.4: ratings are a weak proxy on average, but a strong disambiguator at the text-entropy ceiling. We flag this as an exploratory observation on a small, disagreement-enriched subset, not a general claim.

8.3 Human agreement: the annotator-agreement reference

To bound what "correct" even means on wild short reviews, a 200-row random subset of ood1000 was blind-labelled by a second human annotator. Agreement with the released reference labels is **71.5%** (Cohen's $\kappa = 0.568$, bootstrap 95% CI $[0.47, 0.66]$) — note this is a human-vs-reference agreement, not a pure human–human figure: on this subset the reference labels are 47 human-adjudicated and 153 two-pass LLM-consensus rows. The disagreement is not uniform: on items the first annotator labelled neutral, the second agrees only 49% of the time, relabelling 37 of 80 as positive. Two humans do not share a stable neutral boundary. The instability is even sharper on the genuinely contested rows: within the same 200-row subset, the 47 human-adjudicated rows are exactly those where the two LLM passes disagreed most — passA-vs-passB agreement there is only 29.8% ($\kappa = -0.18$), versus 86.7% overall on ood1000. LLM self-consistency collapses precisely where humans disagree (Davani et al., 2022). We report Cohen's κ throughout following standard IAA practice (Artstein and Poesio, 2008); the documented under-reporting of annotation metadata in NLP motivates our per-row `final_source` provenance field (Kunilovskaya et al., 2026; Al-Khalifa, 2026). Model errors concentrate on exactly these ambiguous rows: the best AL mix (al17k) scores 79.5% on annotator-agreement items vs. 61.7% on disagreement items (v3: 67.8% vs. 46.6%), and AL training lifts neutral recall on disagreement rows from 0.41 to 0.77 (Figure 6).

Meanwhile our best OOD system scores **77.1%** three-class after AL scaling (72.0% with al1000 labels alone) — numerically above the 71.5% annotator-vs-mixed-reference agreement measured on the 200-item subset described above, though the two quantities are not strictly comparable (accuracy vs. pairwise agreement, and the reference side is majority LLM-consensus). The appropriate reading is not "the model is imperfect" but that the model's OOD accuracy is numerically in the same range as measured annotator-vs-reference agreement, though the two metrics are not commensurable; residual error is plausibly dominated by annotator-level ambiguity. For reference, the two LLM annotation passes (same model, different few-shot prompts, blind to each other and to star ratings) agree with each other at 83.5% raw agreement ($\kappa = 0.75$) on the same subset — higher than the 71.5% human annotator-vs-reference figure, but this is single-model self-consistency inflated by shared inductive bias and literal rubric following, not evidence of superior annotation quality; which is why human adjudication, not LLM consensus, anchors the final labels.

8.4 Comparison with a frontier LLM

A natural reviewer question is whether a large general-purpose LLM obviates the whole pipeline. We evaluated DeepSeek-V4.1-Flash (a ∼763B-parameter API model) and DeepSeek-V4-Pro (∼1.6T parameters) zero-shot on the same adjudicated labels, with a compact JSON-only prompt and low reasoning effort.

Table 7: Frontier-LLM vs. local pipeline on lab2000 (API snapshots evaluated November 2026). Parse failures are excluded from accuracy denominators and reported separately. Latency is mean per-request wall time. "—" = metric not reported.

System	Params	ood1000 3-class	ood1000 binary	Latency
DeepSeek-V4-Pro	∼1.6T	0.844 (n=950)	—	∼5.9s API
DeepSeek-V4.1-Flash	∼763B	0.820 (n=992)	**0.976** (n=661)	∼1.7s API
Qwen3-4B (local)	∼4B	0.706 (n=1000)	—	∼2.3s local
Deployed v3	∼1.5B	0.650	0.850	∼0.1s local
SFT+mix (gold+al17k)	∼1.6B	**0.771**	0.923	∼0.1s local

On the full random OOD set DeepSeek-V4.1-Flash leads by ∼5pp (V4-Pro by ∼8pp) — unsurprising

given the orders-of-magnitude parameter gap (∼763B–1.6T vs. our 0.1–1.6B local models) and the corpus exposure our student lacks (on the 992 jointly parseable rows vs. Flash, McNemar $p = 1.7 \times 10^{-3}$). Three qualifications keep the comparison honest. First, the neutral-class wall hits everyone: DeepSeek's own binary accuracy (97.6%) exceeds its three-class accuracy (82.0%) by 15.6pp, the same polarity-vs-neutral gap we document — intrinsic ambiguity is not a small-model artefact. Second, on the human-adjudicated hard subset (195 ood rows jointly parseable by DeepSeek) the gap nearly closes: DeepSeek reaches 59.5% on the ood portion vs. 58.5% for our al1000-era ensemble mix (comparable within noise; the best al17kds mix scores 56.0% on the full 200-row adjudicated subset); the subset is enriched for exactly the cases where rubric-following and world knowledge diverge. Third, the deployment economics differ by orders of magnitude: the local system runs offline on a single consumer GPU at ∼0.1s/review with zero per-token cost and no data leaving the machine, while the API model requires network access, per-call billing, and ships user text to a third party — decisive constraints in privacy-sensitive deployments. We therefore position the contribution not as "beats LLMs" but as "task-calibrated compact model approaching the measured annotator-agreement range, at edge-deployable cost."

8.5 Can LLM pseudo-labels replace human adjudication?

The frontier-LLM result above suggests an obvious scaling path: if the 763B Flash model labels the AL set well, its labels may substitute for expensive human adjudication — LLM-generated pseudo-labels are an established augmentation strategy in their own right (Zenimoto et al., 2024; Neveditsin et al., 2025). We tested this directly — a MacBERT student SFT'd on gold-600 + al1000 with *DeepSeek-V4.1-Flash* labels (987 parseable rows) instead of the released al1000 labels.

Table 8: Released al1000 labels vs. DeepSeek pseudo-labels for AL expansion (ood1000, mixed-provenance reference: 200 human + 800 LLM-consensus rows).

AL labels	3-class acc / F1	Binary acc / F1
gold-600 only	0.626 / 0.611	0.832 / 0.831
gold-600 + al1000 (released)	0.714 / 0.710	**0.880 / 0.875**
gold-600 + al1000 (DS-flash)	**0.713 / 0.714**	0.863 / 0.860

DeepSeek pseudo-labels match the released labels on the three-class task (71.3% vs. 71.4%) and trail slightly on binary (86.3% vs. 88.0%). An ensemble mix at 0.5:0.5 reaches 73.4% (split-half 0.734 ± 0.015), nominally above the al1000-label mix.

Scaling: 1K AL → 5.9K hybrid. This near-equivalence (within noise; 71.3% vs. 71.4%) raises the obvious question — if LLM labels substitute for the released al1000 labels at 1K, do they *scale* where human annotation cannot? We sampled **al5000**: the next 5,000 margin/entropy-selected corpus rows (same band and class quotas as al1000, disjoint from all lab2000 texts), labelled them with DeepSeek-V4.1-Flash (4,906 parseable; label distribution 1,563 pos / 2,552 neu / 791 neg — neutral-heavy as designed), and trained the same SFT recipe. With ∼6.5K AL rows the epoch optimum shifts to 4 (8 epochs overfits; the e4-vs-e8 choice was made on ood1000, a single discrete disclosure-level hyperparameter decision — see Section 9).

Table 9: AL-scaling on ood1000 (mixed-provenance labels: 479 human-adjudicated + two-pass LLM consensus; per-set split in Section 3.5). All rows are gold-600 + listed AL data, MacBERT SFT. "mix" rows are probability averages of v3 and the SFT model; the fusion weight is selected by stratified 5-fold cross-validation (out-of-fold predictions, no row's label influences its own weight). Ranges denote spread across random seeds (five for three-class, four for binary). Binary evaluation excludes neutral examples; "—" = not reported.

AL training data	3-class acc / F1	Binary acc / F1
gold-600 only	0.626 / 0.611	0.832 / 0.831
gold-600 + al1000 (released)	0.714 / 0.710	0.880 / 0.875
gold-600 + al1000 (DS labels)	0.713 / 0.714	0.863 / 0.860
gold-600 + al5000 (DS labels)	0.719 / 0.720	0.893 / 0.890
gold-600 + al6k (all DS labels)	0.725 / 0.727	0.893 / 0.890
gold-600 + al6k (hybrid)	**0.729 / 0.731**	**0.911 / 0.908**
gold-600 + al11k (1K AL + 9.9K DS)	0.734 / 0.737	**0.914 / 0.912**
gold-600 + al11k (1K AL + 4.9K DS + 5K Qwen3-4B)	0.703 / 0.706	0.902 / 0.898
gold-600 + al11k (all DS, no AL core)	0.727 / 0.732	0.900 / 0.898
gold-600 + al3k (disagreement-mined DS)	0.703 / 0.704	0.877 / 0.874
gold-600 + al14k (1K AL + 12.9K DS)	0.744 / 0.747	0.914 / 0.911
gold-600 + al17k (1K AL + 15.9K DS)	0.730–0.746 / 0.732–0.749	0.915–0.924 / 0.912–0.922
+ v3 mix (al6kh)	0.742 / 0.743	0.909 / 0.907
+ v3 mix (al11kds, best of 3 seeds)	0.757 / 0.760	—
+ v3 mix (al11kds 3-seed avg)	0.749 / 0.752	—
+ v3 mix (al11kds binary e8)	—	0.915 / 0.913
+ v3 mix (al14kds, 3 seeds)	0.748–0.757	—
+ v3 mix (al14kds 3-seed avg)	0.755 / 0.758	—
+ v3 mix (al14kds binary)	—	0.909 / 0.907
+ v3 mix (al17kds, 5 seeds)	0.746–0.763	—
+ v3 mix (al17kds 5-seed avg)	**0.771 / 0.774**	—
+ v3 mix (al17kds binary e8 4-seed avg)	—	0.923 / 0.920

Two observations. First, **scale beats provenance**: 5,893 pure-LLM labels (72.5%) already exceed the 1,000-row al1000 set's mixed-provenance labels (71.4%). The teachers' biases differ usefully: DeepSeek under-calls *positive* on ambiguous rows (only 65% agreement with released-label positives, relabelling 156/464 as neutral) while agreeing with the released al1000 labels on neutral at 88% — the human-adjudicated core corrects exactly this polarity bias, which is why the hybrid wins. Second, the **hybrid is best**: keeping the al1000 set's mixed-provenance labels on the hardest 1,000 rows and adding 4,906 LLM labels reaches 72.9% standalone and **74.2% / 0.743** three-class and **90.9% / 0.907** binary after mixing — the strongest OOD numbers at this scale, still ∼7pp below the frontier LLM but at ∼0.1s local latency.

Diminishing returns and teacher quality at scale. We pushed the same protocol to ∼11K AL rows twice: once adding a second 5,000 DeepSeek-labelled batch (al11k-DS), once replacing it with 5,000 labels from a *local* Qwen3-4B teacher scored by label-logprob (al11k-Qwen; the local model reaches only 70.6% on ood1000 zero-shot vs. DeepSeek's 82.0%). The all-API-teacher run edges the hybrid forward (73.4% standalone, 91.5% binary, and the best gold-600 cross-fit of the 11K-scale models at 81.0%), but the three-class mix is statistically flat vs. 6K (74.9% seed-averaged vs. 74.2%, McNemar $p = 0.56$) — the 1K→6K gain does not repeat at 6K→11K, marking the onset of diminishing returns in the ambiguity region. The gain over the 1K-AL mix is nonetheless significant (McNemar $p = 0.041$, 109 vs. 80 discordant pairs), and the mix's OOD calibration improves with it (ECE 0.037 vs. v3's 0.159). The local-teacher run is clearly worse (70.3% standalone, 72.4% mixed, 90.2% standalone binary, cross-fit 79.5%;

pairwise McNemar vs. the DS-teacher mix reaches $p = 0.041$ — teacher quality is a real, measurable gap, not noise): at identical scale, pseudo-label quality — not quantity — is the binding constraint, and a 4B open teacher cannot substitute for the frontier API here. The teachers' disagreement is structured, not random: on the same 5,000 uncertainty-sampled rows their labels agree on only 62.3%, with Qwen over-calling positive (725 of DeepSeek's neutral rows) and under-calling negative (396 of its negative rows relabelled neutral) — a distributional bias the student faithfully inherits. An additional ablation replaces the released al1000 labels with DeepSeek labels at the same 11K scale (al11k-allDS): the three-class mix is unchanged (75.2%, McNemar $p = 0.80$ vs. the al1000-core mix) and binary stays within noise as well (90.3% vs. 91.4%, $p = 0.28$) — at this scale the released human labels show no measurable marginal value over DeepSeek labels on either task. A third round then tests whether *where* AL rows are drawn matters at the margin: we mined 3,000 new rows from the student-disagreement pool (rows where two independently-trained student variants disagree on the argmax label), labelled them with DeepSeek (al3kds), and added them to form al14kds (1K AL + 12.9K DS). The 3K subset alone is strikingly label-efficient — 70.3% standalone, 71.5% mixed, roughly matching the 1K-AL-label mix (72.0%) at 3$\times$ the label count but API-token rather than adjudication cost — while the full 14K mix reaches **75.5%** three-class (74.8–75.7% across three seeds, 75.5% seed-averaged) and **90.9%** binary, with gold-600 cross-fit 81.5%. The difference over the best 11K seed is within noise at n=1000 (McNemar $p = 0.91$), and the 14K mix remains significantly above the 1K-AL mix ($p \approx 0.01$, 107 vs. 72 discordant). A fourth identical round (al17kds, another 3,000 disagreement-mined DeepSeek rows) reaches **77.1%** three-class seed-averaged (74.6–76.3% across five seeds) and **92.3%** binary, with gold-600 cross-fit 81.0% — the best observed numbers, though +1.6pp over 14K is again within noise (McNemar $p = 0.064$), confirming saturation in the $\sim$75–77% band — the honest reading is that disagreement-targeted sampling helps, but the three-class curve is approaching its ceiling. The honest scaling curve is therefore: rapid gains to $\sim$6K AL rows, then saturation on the three-class task while binary and in-domain robustness keep improving. The total API cost for all 15,906 additional DeepSeek labels was a few hundred thousand tokens — orders of magnitude cheaper than equivalent human annotation.

Stratification checks. Two cuts address the obvious objections. *Provenance*: the al17k gain is not an artifact of LLM-consensus labels — on the 800 consensus rows the mix improves over v3 by +12.7pp (69.6%→82.4%) and on the 200 human-adjudicated rows by +9.5pp (46.5%→56.0%). The adjudicated subset is intrinsically harder (author adjudication concentrated on inter-pass disagreements), yet the gain replicates there. The same split also favors the LLM teacher: DeepSeek-Flash scores 87.5% on the 800 consensus rows versus 82.4% for the mix, but only 59.5% (on its 195 parseable adjudicated rows) versus 58.5% for the mix on the 200 adjudicated rows — the consensus labels partially encode the teacher's own annotation conventions, which is why the adjudicated subset is the more informative cut. *Length*: gains concentrate on terse reviews — of the +121 net discordant wins over v3, 58 fall on $\leq$20-character rows — while the $>$100-character slice remains weakest for every system (mix 72.4% vs. v3 61.2% on ood1000; 66.7% on fresh500), consistent with the length-as-domain-signal caveat in the router analysis below. Calibration also improves with scale: OOD ECE falls from v3's 0.159 to 0.063 (al17k) and 0.027 (al14k).

Fresh-sample robustness check. Because AL rows were sampled on corpus uncertainty and mix weights by cross-validation on ood1000, we additionally sampled **fresh500** — 500 uniformly-random corpus reviews with zero *text* overlap against gold-600 and every AL pool through al17k, but only *partial* row-level independence: 70.2% of fresh500 comment IDs appear in at least one corpus-derived training pool, and the deployed corpus neutral detector was trained on 13.2% of fresh500 rows (13.0% of ood1000 rows; Section 6.2). fresh500 is therefore a replication check on a partially-exposed distribution with teacher-generated labels, not a clean holdout — labelled by the same DeepSeek teacher. With weights frozen at the ood1000 cross-fit values (w_{SFT}=0.6 three-class, the modal fold weight; 0.5 binary, the fold median — the e8 binary fold weights are bimodal, 0.45–0.95), the headline al17kds mix scores **77.0%**

three-class vs. v3's 58.8% (McNemar $p = 9.1 \times 10^{-15}$) and **93.7%** binary vs. 86.3% ($p = 1.9 \times 10^{-4}$) — the AL gain replicates on this partially-exposed sample and is not an artifact of benchmark-specific tuning. (The earlier al14k checkpoint scored 74.6% / 92.6% under the same protocol; the standalone al17k SFT student reaches 78.0% five-seed-averaged three-class on fresh500 — the frozen ood1000-tuned mix weights do not transfer perfectly to a new distribution.) (Caveat: fresh500 labels are teacher-generated, not human gold, so these numbers measure generalisation to the teacher's label distribution; binary, being less ambiguous, is less affected by this alignment.)

The practical protocol this validates is exactly the one lab2000 itself used: *LLM-label at scale, human-adjudicate a verification core*. Future work (ii) of Section 10 is thereby partially realised.

Clean human test (human400). We constructed human400 as an *identifier-disjoint, single-annotator, length-stratified* human evaluation set: 400 corpus reviews sampled with length stratification (160/120/80/40 across $\leq$20, 21–50, 51–100, $>$100 characters — a stratified benchmark design, not a corpus-proportional population sample) after excluding every known training and evaluation pool by comment ID, and blind-labelled under written guidelines with star ratings hidden; a post-hoc audit confirmed zero identifier overlap (protocol and guidelines released with the data). Identifier disjointness alone does not prove text-level contamination-free status: a text-level overlap audit (exact, normalized, and near-duplicate Jaccard > 0.8 matching) against all project training pools is reported for this set: exact and near-duplicate overlap with the pools of evaluated systems is 0/400, and normalized matching flags only two ultra-short boilerplate rows in corpus pseudo-label pools feeding the deployed neutral-detector channel (audit script `scripts/audit_human400_textlevel.py`). The set is single-annotator, so it supports a human-reference evaluation but yields no human–human IAA. On it, v3 reaches 55.8% three-class accuracy (macro-F1 55.4) and 76.5% binary accuracy on the 221 non-neutral rows. A star-rating lookup (4–5$\rightarrow$positive, 3$\rightarrow$neutral, 1–2$\rightarrow$negative) achieves *exactly* the same 223/400 correct (55.75%), and a trivial binary rating rule reaches 83.7%, exceeding v3's 76.5%: per-rating, only 62.7% of 5-star and 50.7% of 4-star reviews are human-positive — stars encode recommendation intent, not semantic stance. The frozen-weight al17k mixture reaches **64.8%** three-class (macro-F1 63.5), $+9.0$pp over both v3 and the star lookup (McNemar $p{=}1.3{\times}10^{-3}$ and $7.4{\times}10^{-3}$), with per-seed mixes stable at 63.5–66.0%. The AL gains therefore reflect genuine generalisation to human labels rather than label-convention alignment; the adjudicated-subset pattern (teacher $>$ local mix) does not replicate on clean labels. Residual weaknesses: annotator-flagged mixed reviews drop to 40.0% (vs. v3's 53.3%, $n{=}30$), and absolute accuracy remains modest under a single-annotator reference. On the binary subset the frozen mixture ($w{=}0.5$, four seeds) reaches 83.7% accuracy and 81.3 macro-F1 — matching the rating rule's accuracy with better class balance and significantly above v3 (McNemar $p{=}0.02$). Qwen3-4B zero-shot label-logprob scoring reaches 61.5%/59.0 three-class accuracy/macro-F1, between v3 and the mixture (McNemar $p{=}0.26$).

Deployment trade-off (measured, not assumed). A natural question is whether the OOD-hardened mix should replace v3 in the demo. We tested this honestly: the SFT component was cross-fit on gold-600 (5 folds, each holding out 120 gold rows while training on the rest + all al1000), then mixed with v3's OOF probabilities. The SFT student alone reaches only **80.0% / 0.730** on gold-600 (80.2% / 0.737 for the scaled al6k-hybrid variant, 81.0% / 0.751 for al11k-DS, 81.5% for al14k-DS, 81.0% for al17k-DS; vs. v3's 89.5% / 0.859), and SFT-heavy mixture weights land at 80–82% (lighter weights reach 85.8–88.5%; no weight exceeds v3's 89.5%) — training on terse ambiguous AL examples shifts the decision boundary away from the longer, single-film gold distribution. OOD-hardening and in-domain accuracy are therefore in tension here; we retain v3 as the deployed model and report the mix as an OOD-specialist candidate. A length-aware router (short reviews $\rightarrow$ mix, long $\rightarrow$ v3) traces a Pareto frontier rather than a free lunch: at threshold 20 chars it gives 87.3% gold / 71.0% OOD, between v3 (89.5% / 65.0%) and the pure mix ($\sim$82% / 77.1%). Length is an imperfect domain signal (32% of gold reviews are $<$50 chars;

26% of ood1000 are $\geq$50), so we leave a learned domain router to future work. Table 10 traces the full frontier under a fixed fusion weight (w_{SFT}=0.6, the modal cross-fit value): every threshold buys OOD accuracy with gold accuracy, with no dominant point.

Table 10: Length-router frontier: reviews $\leq T$ chars routed to the AL mix, longer ones to v3 (fixed w_{SFT}=0.6 fusion). Gold-600 and ood1000 three-class accuracy; the pure-mix endpoint's cross-fit OOF estimate is 0.771 (Table 9).

Threshold T	Gold-600	ood1000
0 (pure v3)	0.895	0.650
20	0.873	0.710
50	0.865	0.742
100	0.845	0.760
∞ (pure mix)	0.822	0.774

8.6 What lab2000 does not claim

al1000 is a training set; its headline numbers (e.g. the rating-rule comparison) characterise difficulty, not generalisation. The human-adjudicated 479 rows are a non-random, disagreement-enriched subset; accuracies on it are lower for every system and are reported for the LLM comparison only. LLM-assisted pre-annotations were adjudicated by a single human author plus a 200-row second-human sample; the released data marks per-row provenance (`final_source`) so downstream users can restrict to the 479-row human-verified core if desired.

9 Limitations and Threats to Validity

- **Domain narrowness (mitigated, not eliminated).** The gold set is 600 reviews from a *single film page*; the lab2000 study quantifies the resulting OOD gap (65.0% three-class) and shows 1,000 actively-selected rows (mixed-provenance labels) recover most of it (72.0%). ood1000 is *not* a held-out test, however — it is the development set (next item) — so the recovered 72.0% is a development estimate.
- **ood1000 is a development set, not a test.** Every Stage-2 selection decision — AL scale (1K$\rightarrow$17K), teacher choice (DeepSeek / local Qwen / hybrid), epoch count (e4/e8), seed subsets, $\sim$26 mix configurations over $\sim$73 SFT runs, and the 5-fold cross-fit fusion weights — was made on ood1000 itself. Reported ood1000 numbers are therefore development estimates with an estimated $+0.3$–1.0pp best-of-N optimism, and McNemar p-values computed on the same set are anti-conservative. The identifier-level disjointness check is human400 (Section 8.5): single-annotator, so it supports a human-reference evaluation but yields no human–human IAA; its length-stratified sampling is a benchmark design rather than a natural-distribution population estimate, and a text-level overlap audit is reported in that section.
- **Row-level train/eval overlap.** 68.7% of ood1000, 70.2% of fresh500, and 74.8% of lab500 comment IDs appear in at least one corpus-derived training pool; the deployed corpus neutral detector was trained on 13.0% of ood1000 rows and 13.2% of fresh500 rows. The AL/SFT training pools themselves have zero comment-ID overlap with the test sets, and gold-600 carries no row-level overlap with them apart from eight exact-duplicate text pairs (Leak C, Section 6.2) and a handful of degenerate short texts. lab500 additionally cannot be called "never trained": an earlier cross-validation script used lab500 rows as its own training folds.

- **lab2000 label provenance.** Pre-annotations are LLM-generated; 479/2,000 rows carry human adjudication — collected with the student model's predictions visible in the interface (anchoring risk, Section 3.5) — and a 200-row second-human IAA sample ($\kappa = 0.57$, a blended human-vs-reference figure, not pure human IAA). Consensus labels are reliable but not infallible; per-row `final_source` is released.
- **Pseudo-label ceiling.** Pseudo labels inherit teacher biases; the flywheel saturates after one round (round-2 mean confidence 0.986 = memorisation). The lab2000 results suggest the teacher's narrow-domain prior is the binding constraint on OOD pseudo-labels.
- **Statistical power.** At $n = 600$, differences below $\sim$1pp are unresolvable; the lab2000 OOD study ($n = 1{,}000$) restores power for the headline Stage-2 claims, while gold-set micro-differences remain unresolved.
- **Engineering constraints.** The ROCm/Windows GPU stack required a driver-level reboot during 1.3B feature extraction; inference latency figures are single-GPU measurements.

9.1 Ethics and Data Availability

The Douban corpus derives from a platform whose terms of service do not authorise large-scale collection or model training; we used it for non-commercial research only and claim no permission (Bender and Friedman, 2018). Repository hygiene was enforced by rewriting the git history (2026-11-09): reviewer identifiers (nick, IP address, timestamps, comment/user/movie IDs, vote counts) are stripped from all released records. Following the upstream Moviedata-10M terms, which prohibit redistribution, the $\sim$4,600 benchmark and evaluation rows (gold-600, lab2000, fresh500, lab500) are released as labels, identifiers and SHA-256 text hashes rather than raw text; users may re-join source texts under the upstream license (synthetic augmentation rows remain in full). No reviewer consent was obtained — the texts are short public reviews — and exposure was minimised accordingly; dataset documentation follows datasheet practice (Gebru et al., 2021). Two external-processing disclosures: roughly 18,500 unique review texts were sent to the DeepSeek API for pseudo-labelling — DeepSeek's published privacy policy permits the provider to use de-identified inputs and outputs for service improvement, no zero-retention agreement was in place, and API data is stored in the People's Republic of China — and all 600 gold texts were processed on Alibaba Cloud PAI for the platform comparison in Section 7. Annotation was performed by the authors and one additional annotator on publicly posted, non-sensitive film reviews; users were not contacted, and the task involved no human-subjects experimentation beyond the annotators' own labeling of public posts.

10 Conclusion and Future Work

On a realistic low-resource Chinese sentiment task, corpus-scale pseudo-labelling is the highest-leverage ingredient: it lifts neutral recall from 54% to 71% and lets a single compact student nearly match a multi-model ensemble. Specialist decomposition on corpus-trained neutral detectors yields a positive point estimate and higher neutral-class recall, but the overall Macro-F1 gain is not statistically significant. Star ratings alone are insufficient supervision — measured, not assumed.

The lab2000 study then stress-tests the result honestly. The gold-trained system drops to 65.0% three-class on wild cross-movie reviews; 1,000 actively-selected boundary examples (279 human-adjudicated + 721 LLM-consensus labels) recover the gap to 72.0% (the underlying SFT gain: $+8.8$pp, McNemar $p = 4.0 \times 10^{-8}$), and scaling the same protocol with frontier-LLM pseudo-labels reaches **77.1%** three-class and **92.3%** binary. Three empirical observations emerge: active learning requires a gold anchor, hard labels may outperform soft distillation once the evaluation domain shifts, and star ratings become

the strongest available signal precisely where review text is most ambiguous. Against frontier LLMs of ∼763B–1.6T parameters the compact pipeline trails on raw OOD accuracy but approaches frontier-LLM accuracy on human-adjudicated hard items within noise, at one-to-two orders of magnitude lower latency and cost. The deployed inference pipeline operates locally without external data transfer; however, external APIs were used during the research-stage annotation process.

Future work: (i) scale the adjudicated AL set (the DS-teacher pilot in Section 8.5 tests whether frontier-LLM pseudo-labels can substitute for human adjudication at scale); (ii) use stronger LLM teachers for corpus re-labelling to inject the world knowledge the student lacks — lab2000 shows this is the dominant residual gap; (iii) test MacBERT-large students and the full 271K unbalanced pool; (iv) domain-adversarial or multi-domain training to close the gold→corpus gap without per-domain annotation.

Reproducibility

All scripts, out-of-fold predictions, experiment records, and the lab2000 benchmark data are in the project repository (`https://github.com/shuakami/douban-sentiment`). The 911K corpus is external (research use only; only the ∼4.6K benchmark/evaluation texts are released, with reviewer identifiers stripped, and the active-learning pool's raw text is not redistributed — Section 9.1); trained checkpoints are published at `https://huggingface.co/Shuakami/douban-sentiment-weights`. The full preprocessing → pseudo-label → train → evaluate pipeline is scripted end-to-end. Key artifacts: `out/ens_lite_3class_v3.json` (deployed OOF predictions + probabilities), `out/lab500_*` (OOD study), `data/lab2000/` (benchmark + teacher labels), `scripts/` (all training/eval code), `PAPER.md` (condensed report). Hardware: AMD RX 9070 GRE, ROCm, PyTorch bf16.

A Training Details and Reproducibility

A.1 Hyperparameters

Table 11: Key hyperparameters of the deployed pipeline.

Component	Setting	Value
Distilled student	backbone	`hfl/chinese-macbert-base`
	distill α/τ	0.5 / 2
	epochs / max-len	8 / 256
	folds (OOF)	5
Judge head	backbone	MegatronBERT-1.3B (frozen)
	features	4096-d (CLS + mean-pooled)
	head	2-layer MLP
Neutral detector	backbone	MacBERT-base
	train rows	115,000 (per-class cap 60K, 2 epochs)
Pseudo pool	confidence	max $p \geq 0.7$
	rows	271,905 (56.5K/68.6K/146.8K)
Calibration	method	temperature scaling
	T	0.37 (NLL on gold)
Ensemble weights	dep	student 1 : judge 2
	spec blend	0.5
	p_n mix	0.6 judge-head : 0.4 corpus detector

A.2 Artifacts

- `out/ens_lite_3class_v3.json` — deployed 3-class OOF predictions and probabilities (all 600 rows).
- `out/ens_lite_binary.json` — deployed binary OOF.
- `out/lab500_*` — lab500 pilot: samples, two-pass LLM labels, predictions, cross-validation results.
- `data/reviews_gold_al1000.jsonl` + `data/lab2000/` — lab2000 benchmark: 2,000 adjudicated labels with per-row provenance, dual-pass pre-annotations, DeepSeek/Qwen teacher predictions and labels, fresh500, and all Stage-2 experiment outputs (see `data/lab2000/README.md`).
- `scripts/` — end-to-end pipeline covering corpus preprocessing, pseudo-labelling, corpus-head training, distillation, and ensemble evaluation (full script list in the repository).
- `web/` + `web-ui/` — FastAPI backend and React demo.

A.3 Environment

Python 3.12, PyTorch (ROCm), `transformers`, AMD RX 9070 GRE GPU. All GPU jobs serialized (concurrent jobs crash the ROCm/Windows stack). Model weights: MacBERT students ∼400MB each; MegatronBERT-1.3B shared backbone; total deployed footprint under 4GB. Cold start ≈14s, warm inference ≈0.1s per review.

B Appendix: The lab500 Pilot Study

Before lab2000, a 500-row pilot measured the same OOD question with weaker instrumentation. 500 reviews were sampled uniformly at random from the 911K corpus (gold-deduplicated) and labelled independently by two LLM annotators (a general-purpose assistant model and Qwen3-4B zero-shot (Qwen Team, 2025)); the deployed v3 pipeline scored all rows.

Table 12: lab500 pilot results (superseded by lab2000). "—" = metric not reported.

Evaluator	Reference	n	Accuracy
v3 ensemble	LLM-A labels	500	0.604
Qwen3-4B	LLM-A labels	500	0.662
v3 ensemble	two-pass LLM consensus	331	0.740
v3 binary	LLM-A labels	255	**0.918**
MacBERT CV	held-out labels	—	0.600 OOF

Inter-annotator agreement between the two LLMs was only 66.2%, and a model trained on 400 in-domain labels reproduced held-out labels at just 60% OOF — motivating the larger adjudicated lab2000 benchmark. (A later audit found that this CV script consumed lab500 rows as its own training folds, so lab500 cannot be described as never-trained; Section 9.) Rows the teacher found easy (confidence ≥ 0.7) scored 0.757 for every annotator pair; hard rows scored ≈0.53.

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